

Updated Smart City Configuration

Smart City Operations Network

CSE421 — Cisco Packet Tracer Project

Member 1 ID: **21101017** | Member 2 ID: **22101695**

Base Network: **1.1.0.0/16**

(3rd digit of 21101017 = 1 | 5th digit of 22101695 = 1)

Date: 2026-05-03

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Section 1 — IP Addressing Table

Complete address assignment for every router interface, switch, server, printer, and PC across the six units. All inter-router WAN links use the **1.1.7.0/27** range.

Unit	Device	Interface	IP Address	Subnet Mask	Default Gateway
COH unit — LAN 1.1.0.0/23					
COH	Router COH	Gi0/0 (LAN)	1.1.0.1	255.255.254.0	—
COH	Router COH	Gi0/1 (→ EMP)	1.1.7.9	255.255.255.252	—
COH	Router COH	Se0/3/0 (→ TMC)	1.1.7.1	255.255.255.252	—
COH	Router COH	Se0/3/1 (→ EDU)	1.1.7.5	255.255.255.252	—
COH	DHCP-Server	Fa0	1.1.0.2	255.255.254.0	1.1.0.1
COH	DNS-Server	Fa0	1.1.0.3	255.255.254.0	1.1.0.1
COH	WEB-Server (www.metropolis.city)	Fa0	1.1.0.4	255.255.254.0	1.1.0.1
COH	Printer0	Fa0	1.1.0.5	255.255.254.0	1.1.0.1
COH	MAIL-COH (mail.coh.city)	Fa0	1.1.0.6	255.255.254.0	1.1.0.1
COH	SW-COH	Vlan1	1.1.0.7	255.255.254.0	1.1.0.1
COH	PC10 (DHCP)	Fa0	1.1.0.10	255.255.254.0	1.1.0.1
TMC unit — LAN 1.1.2.0/23 (DHCP pool starts .17, Member 1 ID 21101017)					
TMC	Router TMC	Gi0/0 (LAN)	1.1.2.1	255.255.254.0	—
TMC	Router TMC	Gi0/1 (→ Backbone)	1.1.7.17	255.255.255.248	—
TMC	Router TMC	Se0/3/0 (→ COH)	1.1.7.2	255.255.255.252	—
TMC	Router TMC	Se0/3/1 (→ PUN)	1.1.7.13	255.255.255.252	—
TMC	Printer3	Fa0	1.1.2.2	255.255.254.0	1.1.2.1
TMC	MAIL-TMC (mail.tmc.city)	Fa0	1.1.2.3	255.255.254.0	1.1.2.1
TMC	SW-TMC	Vlan1	1.1.2.4	255.255.254.0	1.1.2.1
TMC	PC4 (DHCP)	Fa0	1.1.2.17	255.255.254.0	1.1.2.1
PUN unit — LAN 1.1.4.0/24 (DHCP pool starts .95, Member 2 ID 22101695)					
PUN	Router PUN	Gi0/0 (LAN)	1.1.4.1	255.255.255.0	—
PUN	Router PUN	Gi0/1 (→ Backbone)	1.1.7.18	255.255.255.248	—
PUN	Router PUN	Se0/3/0 (→ TMC)	1.1.7.14	255.255.255.252	—
PUN	Router PUN	Se0/3/1 (→ EDU)	1.1.7.29	255.255.255.252	—
PUN	PUN Printer	Fa0	1.1.4.2	255.255.255.0	1.1.4.1
PUN	SW-PUN	Vlan1	1.1.4.3	255.255.255.0	1.1.4.1
PUN	PUN PC (DHCP)	Fa0	1.1.4.95	255.255.255.0	1.1.4.1
EDU unit — LAN 1.1.5.0/24					
EDU	Router EDU	Gi0/0 (LAN)	1.1.5.1	255.255.255.0	—
EDU	Router EDU	Se0/3/0 (→ COH)	1.1.7.6	255.255.255.252	—
EDU	Router EDU	Se0/3/1 (→ PUN)	1.1.7.30	255.255.255.252	—
EDU	EDU Printer	Fa0	1.1.5.2	255.255.255.0	1.1.5.1
EDU	SW-EDU	Vlan1	1.1.5.3	255.255.255.0	1.1.5.1

Unit	Device	Interface	IP Address	Subnet Mask	Default Gateway
EDU	EDU PC (DHCP)	Fa0	1.1.5.4	255.255.255.0	1.1.5.1
EMP unit	— LAN 1.1.6.0/25				
EMP	Router EMP	Gi0/0 (LAN)	1.1.6.1	255.255.255.128	—
EMP	Router EMP	Gi0/1 (→ COH)	1.1.7.10	255.255.255.252	—
EMP	Router EMP	Gi0/2 (→ Backbone)	1.1.7.19	255.255.255.248	—
EMP	Router EMP	Se0/3/0 (→ RFS)	1.1.7.25	255.255.255.252	—
EMP	EMP Printer	Fa0	1.1.6.2	255.255.255.128	1.1.6.1
EMP	SW-EMP	Vlan1	1.1.6.3	255.255.255.128	1.1.6.1
EMP	EMP PC (DHCP)	Fa0	1.1.6.4	255.255.255.128	1.1.6.1
RFS unit	— LAN 1.1.6.128/26 (static IPs only, no DHCP)				
RFS	Router RFS	Gi0/0 (LAN)	1.1.6.129	255.255.255.192	—
RFS	Router RFS	Se0/3/0 (→ EMP)	1.1.7.26	255.255.255.252	—
RFS	SW-RFS	Vlan1	1.1.6.133	255.255.255.192	1.1.6.129
RFS	RFS PC1	Fa0	1.1.6.130	255.255.255.192	1.1.6.129
RFS	RFS PC2	Fa0	1.1.6.131	255.255.255.192	1.1.6.129
RFS	RFS Printer	Fa0	1.1.6.132	255.255.255.192	1.1.6.129

Section 2 — Full Router Configuration Commands

Each router's complete configuration as deployed in Cisco Packet Tracer 8.x. Apply each block in **privileged EXEC** mode (after pressing Enter at the console prompt).

2.1 COH Router

Demonstrates: next-hop static route to EDU, RIPv2 with passive-interface (so EDU never advertises COH onward to PUN), static redistribution, DHCP relay (ip helper-address).

```
enable
configure terminal
hostname COH
no ip domain-lookup
interface GigabitEthernet0/0
 ip address 1.1.0.1 255.255.254.0
 ip helper-address 1.1.0.2
 no shutdown
interface GigabitEthernet0/1
 ip address 1.1.7.9 255.255.255.252
 no shutdown
interface Serial0/3/0
 ip address 1.1.7.1 255.255.255.252
 clock rate 64000
 no shutdown
interface Serial0/3/1
 ip address 1.1.7.5 255.255.255.252
 clock rate 64000
 no shutdown
! Next-hop static to EDU LAN
ip route 1.1.5.0 255.255.255.0 1.1.7.6
router rip
 version 2
 no auto-summary
 network 1.1.0.0
 passive-interface Serial0/3/1
 redistribute static metric 3
end
write memory
```

2.2 TMC Router

Demonstrates: RIPv2, router-based DHCP server with pool starting at .17 (Member 1 ID 21101017 last 2 digits), floating static to COH via PUN backbone with AD 150.

```
enable
configure terminal
hostname TMC
no ip domain-lookup
interface GigabitEthernet0/0
  ip address 1.1.2.1 255.255.254.0
  no shutdown
interface GigabitEthernet0/1
  ip address 1.1.7.17 255.255.255.248
  no shutdown
interface Serial0/3/0
  ip address 1.1.7.2 255.255.255.252
  no shutdown
interface Serial0/3/1
  ip address 1.1.7.13 255.255.255.252
  clock rate 64000
  no shutdown
! DHCP - pool starts at .17 (Member 1 ID 21101017 last 2 digits)
ip dhcp excluded-address 1.1.2.1 1.1.2.16
ip dhcp pool TMC_POOL
  network 1.1.2.0 255.255.254.0
  default-router 1.1.2.1
  dns-server 1.1.0.3
router rip
  version 2
  no auto-summary
  network 1.1.0.0
! Floating static to COH LAN via PUN backbone (AD 150)
ip route 1.1.0.0 255.255.254.0 1.1.7.18 150
end
write memory
```

2.3 PUN Router

Demonstrates: RIPv2, router-based DHCP server with pool starting at .95 (Member 2 ID 22101695 last 2 digits), dual-path RIP (direct serial to TMC + backbone /29).

```
enable
configure terminal
hostname PUN
no ip domain-lookup
interface GigabitEthernet0/0
 ip address 1.1.4.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1
 ip address 1.1.7.18 255.255.255.248
 no shutdown
interface Serial0/3/0
 ip address 1.1.7.14 255.255.255.252
 no shutdown
interface Serial0/3/1
 ip address 1.1.7.29 255.255.255.252
 clock rate 64000
 no shutdown
! DHCP - pool starts at .95 (Member 2 ID 22101695 last 2 digits)
ip dhcp excluded-address 1.1.4.1 1.1.4.94
ip dhcp pool PUN_POOL
 network 1.1.4.0 255.255.255.0
 default-router 1.1.4.1
 dns-server 1.1.0.3
router rip
 version 2
 no auto-summary
 network 1.1.0.0
end
write memory
```

2.4 EDU Router

Demonstrates: RIPv2, router-based DHCP server. EDU never re-advertises COH to PUN because COH's Se0/3/1 is passive.

```
enable
configure terminal
hostname EDU
no ip domain-lookup
interface GigabitEthernet0/0
  ip address 1.1.5.1 255.255.255.0
  no shutdown
interface Serial0/3/0
  ip address 1.1.7.6 255.255.255.252
  no shutdown
interface Serial0/3/1
  ip address 1.1.7.30 255.255.255.252
  no shutdown
ip dhcp excluded-address 1.1.5.1 1.1.5.2
ip dhcp pool EDU_POOL
  network 1.1.5.0 255.255.255.0
  default-router 1.1.5.1
  dns-server 1.1.0.3
router rip
  version 2
  no auto-summary
  network 1.1.0.0
end
write memory
```

2.5 EMP Router

Demonstrates: RIPv2, router-based DHCP server, recursive static to COH via TMC's backbone IP, next-hop static to RFS LAN.

```
enable
configure terminal
hostname EMP
no ip domain-lookup
interface GigabitEthernet0/0
  ip address 1.1.6.1 255.255.255.128
  no shutdown
interface GigabitEthernet0/1
  ip address 1.1.7.10 255.255.255.252
  no shutdown
interface GigabitEthernet0/2
  ip address 1.1.7.19 255.255.255.248
  no shutdown
interface Serial0/3/0
  ip address 1.1.7.25 255.255.255.252
  clock rate 64000
  no shutdown
ip dhcp excluded-address 1.1.6.1 1.1.6.2
ip dhcp pool EMP_POOL
  network 1.1.6.0 255.255.255.128
  default-router 1.1.6.1
  dns-server 1.1.0.3
router rip
  version 2
  no auto-summary
  network 1.1.0.0
  ! Recursive static to COH LAN via TMC backbone IP
ip route 1.1.0.0 255.255.254.0 1.1.7.17
  ! Next-hop static to RFS LAN
ip route 1.1.6.128 255.255.255.192 1.1.7.26
end
write memory
```


2.6 RFS Router

Demonstrates: exit-interface static (supernet) covering all other networks via EMP. Static-only - no DHCP, no RIPv2, isolated outpost.

```
enable
configure terminal
hostname RFS
no ip domain-lookup
interface GigabitEthernet0/0
  ip address 1.1.6.129 255.255.255.192
  no shutdown
interface Serial0/3/0
  ip address 1.1.7.26 255.255.255.252
  no shutdown
! Single exit-interface supernet - covers everything via EMP
ip route 1.1.0.0 255.255.0.0 Serial0/3/0
end
write memory
```

2.7 Switch CLI (template for SW-COH, SW-TMC, SW-PUN, SW-EDU, SW-EMP, SW-RFS)

Replace hostname, Vlan1 IP/mask, and default-gateway per the table below. Same block applies to every switch.

```
enable
configure terminal
hostname SW-COH
no ip domain-lookup
interface Vlan1
  ip address 1.1.0.7 255.255.254.0
  no shutdown
ip default-gateway 1.1.0.1
end
write memory
```

Switch	Vlan1 IP	Subnet Mask	Default-Gateway
SW-COH	1.1.0.7	255.255.254.0	1.1.0.1
SW-TMC	1.1.2.4	255.255.254.0	1.1.2.1
SW-PUN	1.1.4.3	255.255.255.0	1.1.4.1
SW-EDU	1.1.5.3	255.255.255.0	1.1.5.1
SW-EMP	1.1.6.3	255.255.255.128	1.1.6.1
SW-RFS	1.1.6.133	255.255.255.192	1.1.6.129

Section 3 — Server Configuration Steps

All servers are **Server-PT** devices configured through the Packet Tracer GUI (no CLI). The COH unit hosts every infrastructure server; MAIL-TMC sits on the TMC LAN.

3.1 DHCP-Server (1.1.0.2) - serves COH LAN only

Step A - IP Configuration (Desktop tab -> IP Configuration)

Choose **Static**, then enter:

IPv4 Address : 1.1.0.2
Subnet Mask : 255.255.254.0
Default Gateway : 1.1.0.1
DNS Server : 1.1.0.3

Step B - DHCP Service (Services tab -> DHCP)

1. Service: **On**
2. Pool Name: **COH_POOL**
3. Default Gateway: **1.1.0.1**
4. DNS Server: **1.1.0.3**
5. Start IP Address: **1.1.0.10**
6. Subnet Mask: **255.255.254.0**
7. Maximum Users: **500**
8. Click **Add**, then **Save**.
9. Click the built-in **serverPool** row -> set Default Gateway = **1.1.0.1**, DNS = **1.1.0.3** -> **Save**. (serverPool cannot be deleted; aligning it prevents an accidental 0.0.0.0 gateway lease.)

Reserved/static (excluded by Start = .10): .1 router, .2 DHCP, .3 DNS, .4 WEB, .5 Printer0, .6 MAIL-COH, .7 SW-COH.

3.2 DNS-Server (1.1.0.3)

Step A - IP Configuration

IPv4 Address : 1.1.0.3
Subnet Mask : 255.255.254.0
Default Gateway : 1.1.0.1
DNS Server : 1.1.0.3

Step B - DNS Service (Services -> DNS)

DNS Service: **On**. For each row below set Type = **A Record**, fill Name and Address, click **Add**:

Name	Type	Address
www.metropolis.city	A Record	1.1.0.4
coh.city	A Record	1.1.0.6
mail.coh.city	A Record	1.1.0.6
tmc.city	A Record	1.1.2.3
mail.tmc.city	A Record	1.1.2.3

The bare-domain records coh.city and tmc.city are required for server-to-server SMTP relay; without them, PC4 -> PC10 mail bounces with 'Domain name not found'.

3.3 WEB-Server (1.1.0.4) - www.metropolis.city

Step A - IP Configuration

IPv4 Address : 1.1.0.4
Subnet Mask : 255.255.254.0

Default Gateway : 1.1.0.1
DNS Server : 1.1.0.3

Step B - HTTP Service (Services -> HTTP)

HTTP: **On**, HTTPS: **On**. Click the **index.html** row in the file list and replace its contents with:

```
<html>
  <center>
    <h1>Smart City Operations - Online.</h1>
  </center>
</html>
```

Click **Save**.

Verification

From any PC -> Desktop -> Web Browser -> enter **http://www.metropolis.city**. Page loads with the above message.

3.4 MAIL-COH (1.1.0.6) - mail.coh.city

Step A - IP Configuration

IPv4 Address : 1.1.0.6
Subnet Mask : 255.255.254.0
Default Gateway : 1.1.0.1
DNS Server : 1.1.0.3

Step B - EMAIL Service (Services -> EMAIL)

1. SMTP Service: **On**
2. POP3 Service: **On**
3. Domain Name: **coh.city** -> click **Set**
4. User Setup -> User: **admin**, Password: **admin123** -> click **+**

3.5 MAIL-TMC (1.1.2.3) - mail.tmc.city

Step A - IP Configuration

IPv4 Address : 1.1.2.3
Subnet Mask : 255.255.254.0
Default Gateway : 1.1.2.1
DNS Server : 1.1.0.3

Step B - EMAIL Service (Services -> EMAIL)

1. SMTP Service: **On**
2. POP3 Service: **On**
3. Domain Name: **tmc.city** -> click **Set**
4. User Setup -> User: **traffic**, Password: **traffic123** -> click **+**

3.6 PC Mail Client Configuration

On each PC -> Desktop -> Email -> Configure Mail:

Field	PC10 (COH)	PC4 (TMC)
Your Name	admin	traffic
Email Address	admin@coh.city	traffic@tmc.city
Incoming Server	mail.coh.city	mail.tmc.city
Outgoing Server	mail.coh.city	mail.tmc.city
User Name	admin	traffic
Password	admin123	traffic123

3.7 End-to-End Mail Verification

1. **PC10** -> Compose -> To: **traffic@tmc.city** -> Subject: 'Hello from COH' -> Send. Bottom pane shows *Send Success*.
2. **PC4** -> Receive. Mail appears in inbox.
3. **PC4** -> Compose -> To: **admin@coh.city** -> Send.
4. **PC10** -> Receive. Mail appears.

End of submission